Improved Performance of a CA-SSL-based Daily Eating Sounds Recognition Model

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***Abstract*—Daily conversation and eating behaviors are crucial for maintaining physical and mental health among older adults. To facilitate monitoring of these behaviors, skin-contact microphones are used, which are useful in recognizing chewing and swallowing behaviors, but the data recorded by these microphones are limited and are seldom labeled. In this study, we focused on the CA-SSL (Class-Agnostic Semi-Supervised Learning) framework used in image recognition for highly accurate recognition of daily eating sounds, which are eating sounds including speech, and proposed a learning method that effectively utilizes a large amount of unlabeled eating sounds collected as training data. We used 1-class labeled eating sounds—which were created from a large amount of unlabeled eating sounds data by adopting a naive pretrained labeling model—for the first fine-tuning of the recognition model of speech and eating sounds. Subsequently, the model was fine-tuned again with manually assigned labeled data to develop the daily eating sounds recognition model. The experimental results showed that the proposed model improved the recognition accuracy of chewing sounds by 1.2 times and swallowing sounds by 22 times compared with the baseline model before applying the 1-class label in the evaluation of simulated daily eating sounds by F1-score.**

Keywords*—*daily eating sounds recognition, skin-contact microphone, CA-SSL, WavLM, machine learning

I. Introduction

The behaviors of speaking and eating are important health indicators for older adults, and research is seeking to monitor these behaviors more accurately. For speech recognition, as open-source high-performance recognition models are available, various technologies have been proposed, such as a speech recognition model for patients with dysarthria and older adults using the SSL (Self-Supervised-Learning) model [1], and real-time speech recognition for multiple speakers has been developed using whisper, a large-scale speech recognition model [2]. With respect to the recognition of eating sounds, methods exist of detecting swallowing from video data with machine learning [3] and to classify the swallowing of food and water using swallowing sounds [4]. A previous study on speech and eating sounds recognition collected speech data during eating and used SVM (Support Vector Machine) to recognize whether a person was eating while speaking and what they were eating [5]. In other methods, a skin-contact microphone is used to record the wearer's speech and eating sounds and limit external noise. A method of detecting intervals between speech and eating sounds has been proposed. This method utilizes data recorded simultaneously using an ordinary microphone and a skin-contact microphone [6]. Additionally, a method for classifying speech, chewing, food swallowing, and water swallowing has been developed. This approach employs a skin-contact microphone and a smartphone [7].

However, due to differences in acoustic characteristics, the sound data recorded by skin-contact microphones are difficult to recognize with recognition models of the sound data recorded by ordinary microphones. Therefore, it is important to create new models that fit the characteristics of these data. However, large datasets of speech and eating sounds recorded using skin-contact microphones are not publicly available. In addition, the process of labeling the collected speech and eating sounds requires substantial time and manpower. For these reasons, it is difficult to prepare a sufficient amount of labeled data, and the lack of training data is a major challenge in the development of new recognition models. To address this issue, a speech recognition model developed for skin-contact microphones using SSL has been proposed for speech recognition [8]. In addition, to improve the accuracy of eating sounds recognition, a data expansion method utilizing n-grams based on the features of eating sounds has been proposed [9]. Such approaches make effective use of small amounts of labeled data to develop highly accurate recognition models for skin-contact microphones.

This study aims to solve the problem of insufficient training data by employing unlabeled data. We have an environment in which speech and eating sounds can be recorded using skin-contact microphones in a soundproof room and by recruiting participants. Therefore, unlabeled data can be easily collected in large quantities, and we can expect to be in a position to develop a highly accurate recognition model by effectively using the recorded unlabeled data as training data.

For the use of unlabeled data, we focus on the CA-SSL (Class-Agnostic Semi-Supervised Learning) [10] method. This is a proposed framework for image recognition that involves two types of models: the labeling model and the image recognition-model. In the labeling model, pre-trained, class-agnostic 1-class labels is assigned to unlabeled images, and these data are used for initial training of the image recognition model. Next, this image recognition model is refined using manually labeled images to develop a highly accurate model. The method shows that initial training with 1-class labeled data

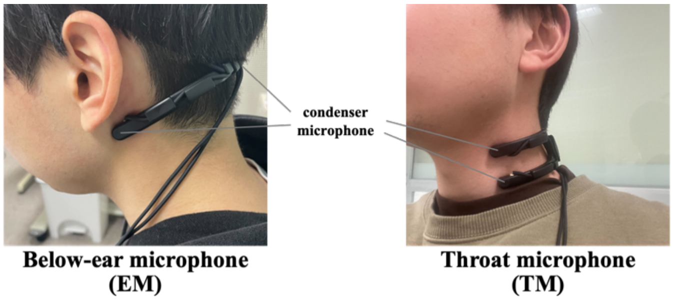

Fig. 1. A below-ear microphone (EM) and a throat microphone (TM). (*The EM has left and right channels, and the TM has upper and lower channels above and below the throat.*)

improves object detection accuracy, which in turn improves image recognition accuracy. In the field of sounds, as object detection corresponds to acoustic detection, it is expected that improved of acoustic detection accuracy will lead to improved accuracy of the recognition of speech recognition and eating sounds. Eating sounds are difficult to recognize because they are more similar to noise than speech. Furthermore, swallowing sounds are more difficult to recognize because they occur less frequently during a meal compared with chewing sounds, resulting in even lower recognition accuracy. Therefore, the application of CA-SSL may improve the recognition of eating sounds without being affected by the bias in the amount of data per class.

Therefore, we propose a method to effectively utilize unlabeled eating sounds based on the CA-SSL framework to improve the recognition accuracy of daily eating sounds mixed with speech sounds. First, unlabeled eating sounds were assigned tentative labels with a labeling model trained on labeled speech and eating sounds. Thus, we trained a recognition model using daily eating sounds by changing the tentative labels to 1-class labels and ultimately developed a highly accurate recognition model by fine-tuning it using the labeled speech and eating sounds. We used our dataset of recorded speech and eating sounds, and pretrained SSL models with English speech were used for the labeling model and the daily eating sounds recognition model.

II. Approach

A. Data Overview

All data were recorded simultaneously using a below-ear microphone (EM) and a throat microphone (TM). These microphones are shown in Fig. 1. The TM is suitable for recording speech and swallowing sounds but not for recording chewing sounds. In contrast, the EM can clearly distinguish chewing sounds while also recording speech and swallowing sounds to some extent. We simultaneously recorded speech and eating sounds of participants wearing both microphones at 44 kHz. We downsampled these data to 16 kHz, a rate that is suitable for inputs for a speech recognition model. Since the skin-contact microphones generate noise in a low-frequency band owing to blood circulation in the neck, the data were passed through a high-pass filter at 100 Hz so that unnecessary low-frequency components could be removed. For each eating sound, we manually prepared strong labels with accurate temporal information for chewing and swallowing. Because accurate temporal information is not necessary for training speech recognition models, weak labels were generated by removing temporal information from the strong labels.

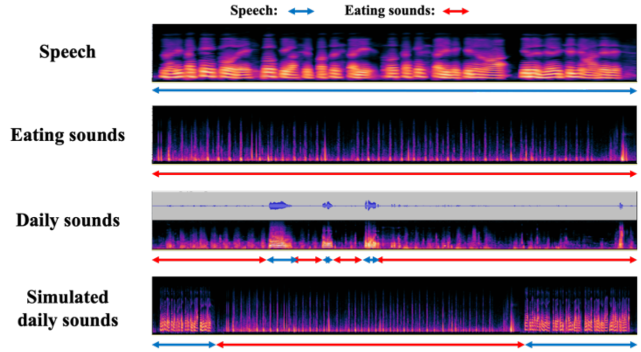

Fig. 2. Four types of data related to this study. (*Note that the eating sounds included in daily sounds are defined as "daily eating sounds" and the eating sounds included in simulated daily sounds are defined as "simulated daily eating sounds."*)

To prevent confusion caused by the naming of data, names were defined in relation to the types of sound contained in one sound data, as shown in Fig. 2. The sounds that contain only speech and those containing only chewing and swallowing sounds were defined as "speech" and "eating sounds." Furthermore, the sounds that include both speech and eating sounds recorded in realistic meal settings were defined as "daily sounds," and the eating sounds included in them were called "daily eating sounds." However, daily sounds are data in which speech and eating sounds overlap in complex ways, and recognizing daily eating sounds is a difficult task as eating sounds are closer to noise than speech. To address this, we prepared "simulated daily sounds," which are artificially created by combining separately recorded speech and eating sounds. As with the daily eating sounds, the eating sounds that are included in the simulated daily sounds are defined as "simulated daily eating sounds."

As a preliminary attempt at the recognition of daily eating sounds, this study used "speech" and "eating sounds" as training data and "eating sounds" and "simulated daily eating sounds," which simplified the task, as evaluation data.

B. Base Fine-tuning Method for Speech and Eating Sounds

To build a model to recognize daily eating sounds, the SSL model WavLM [11] was selected as the base model. To fine-tune WavLM using manually pre-labeled speech and eating sounds, it is necessary to give tokens to the chewing and swallowing sounds. We prepared <C> for chewing and <S> for swallowing tokens, as well as <bos> and <eos> for sentence beginnings and endings, <unk> for unknown events, and <blank> for blank spaces. These were added to the Japanese tokens in hiragana and kanji, and <1-class>, a token representing the 1-class label used in this study, was added to 1,288 tokens for classification. Unless otherwise specified, during the fine-tuning, the parameters of the feature extractor and the transformer were updated to minimize Connectionist Temporal Classification (CTC) loss with the correct labels in the linear layer. The speech recognition library Hugging Face [12] was used to train and evaluate the model.

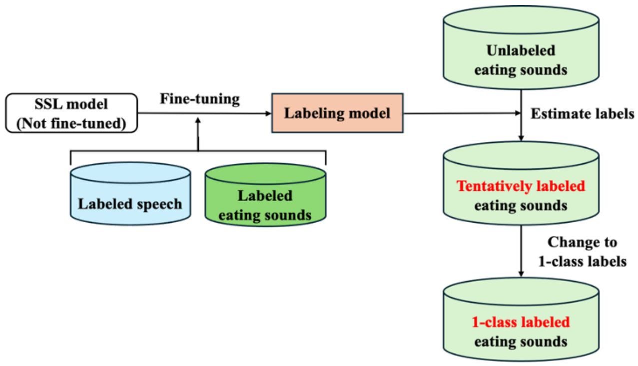

Fig. 3. Procedure for creating 1-class labeled eating sounds

C. *Development of the Proposed Model*

1) Creation of 1-class Labeled Eating Sounds: In this study, a model for daily eating sounds recognition is constructed using CA-SSL [10]. The method of creating 1-class labels is shown in Fig. 3. The daily eating sounds recognition model is constructed by fine-tuning WavLM [11] with labeled speech and eating sounds. With this labeling model, the labels that are estimated with unlabeled eating sounds are used as tentative labels. These labels are all represented by chewing, swallowing, and Japanese tokens, as shown in Section A. To remove classes from these tentative labels, all labels were converted to a 1-class label, <1-class>. In this way, 1-class labeled eating sounds were created.

2) Fine-tuning of the Daily Eating Sounds Recognition Model: Here, a two-step fine-tuning process is used to construct the model. Fig. 4 presents the method of fine-tuning of the daily eating sounds recognition model developed in this study. First, the initial fine-tuning is performed using 1-class labeled eating sounds for WavLM that have not been fine-tuned. Because all 1-class labels are the same, we anticipate that updating the parameters of the transformer section could lead to a loss of the linguistic information obtained during the pre-training. Therefore, we only updated the parameters of the feature extractor. Next, the second fine-tuning is conducted on the constructed model with labeled speech and eating sounds. For this step, we used ground-truth labels instead of 1-class labels. To adapt the model to the task of recognizing speech and eating sounds, all parameters of the feature extractor and of the transformer were updated.

III. Experiment

A. *Experimental Conditions*

1) Model: In this study, two baseline models were constructed in addition to the proposed model, and their recognition performance was evaluated. The proposed model was developed using the method shown in Section II. The baseline 1 model is a fine-tuned model in which speech and eating sounds are labeled without the first fine-tuning shown in Fig. 4. The baseline 2 model was constructed by changing the tuning data in the first fine-tuning from 1-class labeled eating sounds to tentatively labeled eating sounds. In addition, because CA-SSL consists of two steps, namely, “creation of 1-class labels using the labeling model” and “training of the daily eating sounds recognition model,” it is possible to develop a recursive model using the constructed daily eating sounds recognition model as the labeling model and performing this method again. Therefore, we compare the accuracy of recognition of daily eating sounds between the proposed model, which has had this method applied once, and the recognition model, to which this method was applied recursively. The iteration 2 model is that in which the proposed model has been used as the labeling model and applied to the method again, and the iteration 3 model refers to the model in which the method is applied again using the iteration 2 model as the labeling model. Note that WavLM-Large, which was pretrained with 94,000 hours of English speech, was used as the pretraining model for WavLM. To train each model, we targeted the correct weak label in each dataset: 10 % of the training dataset was allocated for validation and training continued until the loss converged.

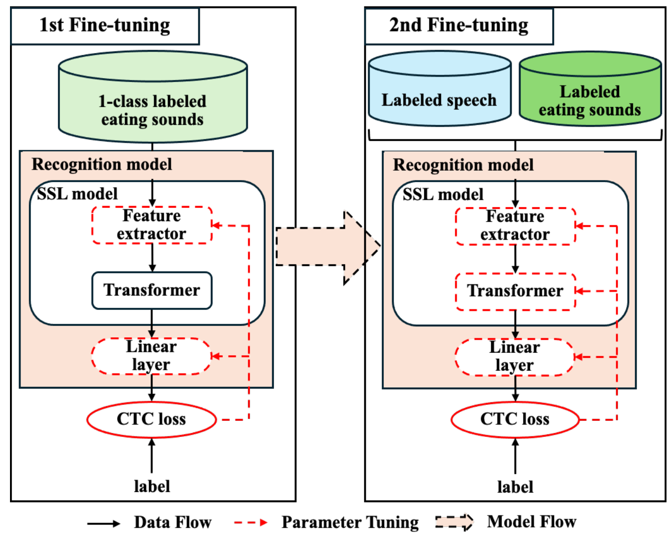

Fig. 4. Fine-tuning process for the daily eating sounds recognition model

2) Dataset: In this experiment, to investigate differences in the model’s recognition accuracy resulting from the differing data characteristics of the two skin-contact microphones, we employed data from the left channel of the EM and the upper channel of the TM. The details of the tuning data are provided in TABLE I. Labeled speech, labeled eating sounds, tentative labeled eating sounds, and 1-class labeled eating sounds were used for tuning data. The labeled speech featured phonemically balanced speech with labels. Labeled eating sounds consisted of eating behaviors involving cabbage, chewing gum, and crackers. Tentatively labeled eating sounds and 1-class labeled eating sounds data included sounds of eating corned beef, in addition to the abovementioned foods, and drinking water. Furthermore, a small amount of the participants' speech was recorded. Two types of data were prepared for evaluation. The first involved eating sounds (five participants, 1 hour), using the same foods as those used to generate the labeled eating foods; this process was followed to evaluate the recognition of these specific foods. The second type of data was simulated daily eating sounds. These data included eating sounds and speech to approximate the sounds heard during everyday meals. The data

TABLE Ⅰ. DETAILS OF TUNING DATA. [a]

Tuning data	Label	Participants	Size(hours)
Labeled speech	Ground truth	40	30
Labeled eating sounds [b]	Ground truth	24	4.5
Tentatively labeled eating sounds	Tentative labels	55	32
1-class labeled eating sounds	1-class labels		

[a.] All data were recorded simultaneously using a below-ear microphone and a throat microphone.

[b.] Although not shown in the table, we used 10× labeled eating sounds, which were generated by applying the augmentation method [9] to the labeled eating sounds.

of 1.5 hours was created by combining the evaluation data for recognition of eating sounds and speech data from reading newspaper articles (five participants, 30 minutes). Participants whose data were evaluated were not included in data tuning.

3) Evaluation Metrics: The character error rate (CER) and F1-score were used as evaluation indices for eating sounds recognition. CER is a commonly used evaluation index in speech recognition models. We applied it to the task in this study by employing weak labels that represent chewing and swallowing sounds with the single character tokens <C> and <S>, respectively. CER allows us to evaluate whether the model is correctly recognizing chewing and swallowing sounds, as well as the number of chewings. However, it is difficult to correctly evaluate the recognition accuracy of swallowing using CER alone because the majority of eating behavior is chewing. Thus, we evaluated the detection performance of each event using the F1-score. The CTC used for the output of the model in this study has the property of compressing the output to a few frames somewhere near the interval rather than to all frames of the estimated event. Therefore, in this experiment, we evaluated the event as correctly detected when the output was included in the ground truth, which is labeled on a frame-by-frame basis. Note that one frame is defined as 20 ms.

B. Experimental Results

The results of recognition performance for each model are shown in TABLE II. TABLE II shows that the proposed model involves superior recognition accuracy to that of the other models in terms of both the CER and F1-score. In particular, the F1-score for the swallowing sound is 2.2 times greater than that of baseline 1, increasing from 0.33 to 0.73. Since there are more and a greater variety of instances of swallowing in the unlabeled eating sounds than in the labeled ones, recognition performance was significantly improved. This suggests that the acoustic features of various swallowing sounds, which could not be obtained only by tenfold data expansion, can be appropriately learned. For chewing sounds, the F1-score increased 1.53-fold from 0.52 to 0.80 compared to baseline 1, indicating that the chewing sounds in the evaluation data were also detected appropriately. As was the case with swallowing sounds, the performance improvement may be due to the fact that the diversity in the patterns and frequency of the chewing sounds increased considerably because many acoustic features of the chewing sounds could be learned from the 1-class labeled

TABLE Ⅱ. CER AND F1-SCORE FOR THE EATING SOUNDS RECOGNITION IN EACH MODEL. [a]

Models	Amount of data used for tuning	CER↓ [eating]	F1-score↑ [chewing]	F1-score↑ [swallowing]
Baseline 1	4.5	14.47	0.52	0.33
Baseline 2(using tentative labels)	32 + 4.5	10.52	0.68	0.22
Proposed	32 + 4.5	**5.61**	**0.80**	**0.73**

[a.] Evaluation data compared 1 hour of the eating sounds of cabbage, chewing gum, and crackers by five participants.

eating sounds. However, this result may be attributable to differences in the amount of data.

Therefore, we compare the proposed model with baseline 2, which has the same amount of data, and the first fine-tuning condition of the proposed model was changed. Relative to baseline 2, the CER shows a 47% decrease. This indicates that learning after changing to the 1-class label was adequate, although the amount of data remained the same. Conversely, comparing baseline 2 to baseline 1, the F1-socre for chewing sounds improved, but the F1-score for swallowing sounds decreased significantly. This may be because the labeling model misidentified swallowing sounds and certain utterances as chewing sounds for the tentative label, which was the label before the change to the 1-class label, producing a recognition model that was not adapted to swallowing sounds. In the proposed method with 1-class labels, all labels are the same and the model learns acoustic features uniformly, regardless of the class of speech, chewing sounds, or swallowing sounds. This difference influences recognition performance. In conclusion, this method is effective for improving the performance of recognition of eating sounds.

Next, we examine differences in the results of food behavior sounds recognition between the EM and TM data. The results are shown in TABLE III. The F1-score for chewing decreased by 0.02 and that for swallowing increased by 0.02 relative to the EM data. This may be attributable to the characteristics of TM data, which are more suitable for recording swallowing sounds than chewing sounds. CER increased by 1.2 times relative to EM data. This is because the number of sounds included in these data is higher for chewing sounds than for swallowing sounds, and in CER, where the number of outputs affects the recognition accuracy, the data for EM, which can output chewing sounds appropriately, are limited.

Furthemore, TABLE IV presents the results for the recognition of daily eating sounds using simulated data for these sounds. The results show that the recognition performance of the proposed model is the best even for the daily eating sounds, which are mixed with speech data. This may be due to the effect of the transition between speech and eating sounds that are included in the daily eating sounds data. Chewing sounds after speech and swallowing sounds before speech are misrecognized because these sounds more closely resemble noise than speech. Conversely, in the proposed model, the F1-score for swallowing sounds decreased from 0.73 to 0.70 for the normal and daily eating sounds data, but the decrease was only a quarter of 0.03, relative to 0.14 in Baseline 1. This

TABLE III. RESULTS OF THE CER AND F1-SCORE FOR EATING SOUNDS RECOGNITION IN EM AND TM DATA.

Models	**CER↓ [eating]**		**F1-score↑ [chewing]**		**F1-score↑ [swallowing]**	
	EM	**TM**	**EM**	**TM**	**EM**	**TM**
Proposed	**5.61**	6.79	**0.80**	0.78	0.73	**0.75**

TABLE IV. CER AND F1-SCORE FOR THE DAILY EATING SOUNDS RECOGNITION IN EACH MODEL. [a]

Models	**Amount of data used for tuning**	**CER↓ [eating]**	**F1-score↑ [chewing]**	**F1-score↑ [swallowing]**
Baseline 1	4.5	16.14	0.49	0.21
Baseline 2 (Using tentative labels)	32+4.5	11.58	0.60	0.08
Proposed	32+4.5	**7.43**	**0.75**	**0.70**

a. The evaluation data consisted of 1.5 hours of simulated daily eating sounds data created by combining the 1-hour eating sounds used in Table 2 with 30 minutes of speech from another five participants.

may be because the model using 1-class labels was able to learn the acoustic features of the transition from swallowing sounds to speech without relying on labels. This indicates that the proposed method can improve the performance of daily eating sounds recognition with mixed speech.

Finally, Fig. 5 presents a comparison of the recognition accuracy of daily eating sounds with a model in which CA-SSL is recursively applied. For both chewing and swallowing sounds, the iteration 3 with three iterations of the proposed method obtained the highest F1-score. Compared with the proposed iteration 1, the F1-score increased by 0.06 for chewing sounds and by 0.05 for swallowing sounds. This indicates that recognition accuracy can be improved through the recursive application of the proposed method. On the other hand, there was a 0.04 increase in the chewing sounds from iteration 1 to iteration 2, but only a 0.02 increase from iteration 2 to iteration 3. In the F1-score of the swallowing sound, it shows the same value in iteration 2 and iteration 3. This suggests that recognition accuracy does not improve in terms of the amount of recursive execution of this method; the rate of increase gradually decreases, and the recognition accuracy converges.

IV. CONCLUSION

In this study, we focused on the CA-SSL framework to improve the performance of the recognition of eating sounds in the daily eating sounds recognition model. 1-class labeled eating sounds were created using the unlabeled eating sounds and were adopted for the first fine-tuning of WavLM. Relative to the existing models and models that were constructed using tentatively labeled eating sounds, the proposed model adequately learns the acoustic features of various eating sounds, showing a significant improvement in recognition performance. It was found that the proposed method contributed significantly to improving the recognition accuracy of swallowing sounds in daily eating sounds recognition. Therefore, further improvement

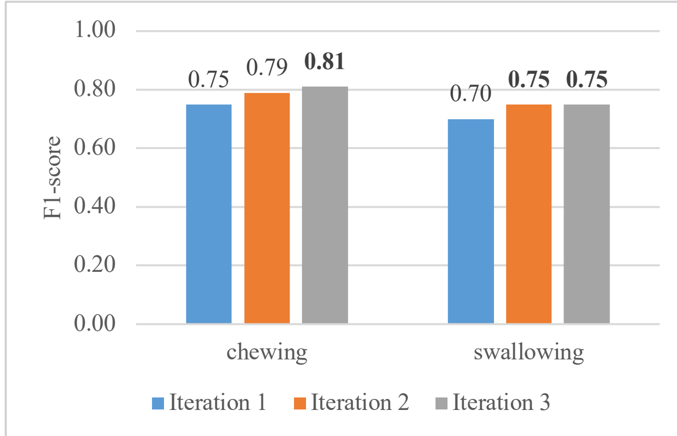

Fig. 5. Comparison of the performance of daily eating sounds recognition models for each time the proposed method is applied. (*Iteration 2 and 3 use 1-class labeled eating sounds, created using the previous iteration as the labeling model, for refining the first fine-tuning.*)

in recognition accuracy was observed when the method was applied recurrently.

In future research, we plan to study methods to improve the eating sounds recognition accuracy of the model, such as new data expansion methods, and to evaluate the model using real-world daily sounds data instead of simulated data.

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